

E-commerce Order Tracker

What it does

This workflow automatically monitors your Email inbox for order confirmation emails, extracts structured order data using AI, checks for duplicates, and logs each order as a row in Google Sheets , giving you a real-time spend tracker across all stores.

How it works (node by node)

1. **Gmail Trigger** — Polls inbox using "Message Received" mode either every 15 minutes or every minute depends on the use case.
2. **Switch node** — Filters emails by subject keywords using case-insensitive matching . Depending on the sub of the email. Routes order-related emails to the Aggregate node. Non-matching emails are dropped via the fallback output.
3. **Aggregate node** — Collects all matching emails into one list before passing downstream. This ensures one AI call processes all emails together instead of one call per email ,reducing cost and improving consistency.
4. **AI Agent (Google Gemini)** — Receives the aggregated email list and extracts structured data from each email body. Uses a system prompt to train it what we exactly need it to extract from the emails and pass the output
5. **Structured Output Parser** — Ensures the AI output to be in a specific format, the response into a defined schema with fields: store, order id, items, amount, currency, order date, delivery date, category, ai reasoning.
6. **Append or Update (Google Sheets)** —Writes one row per order with all extracted fields including timestamp if it finds the required input from the previous nodes and the column to match is order ID to prevent any duplicate order from updating on the sheet

How to run it

1. Connect your Gmail and Google Sheets accounts n8n credentials
2. Create a Google Sheet with these headers in Row 1: Timestamp, Email ID, Store, Order ID, Items, Amount, Currency, Order Date, Delivery Date, Category, AI Reasoning
3. Add your Gemini API key to the Google Gemini Chat Model node
4. In the Google Sheets node set Operation to "Append or Update Row" and Column to Match On to "Order ID"
5. Send yourself a test email with subject containing "order confirmed" and order details in the body
6. Ensure N8n runs the workflow successfully
7. Open your Google Sheet and confirm a new row was added
8. Send the same email again to verify duplicate detection — the existing row should update/overwrite, not create a new one

Gmail Trigger → Switch → Aggregate → AI Agent → Append or Update Row

Point to Note

- Switch node only matches 3 subject patterns — unusual subject lines from some stores may be missed
- AI extraction accuracy depends on how structured the email body is if it is a heavily formatted or detailed email it may confuse the parser
- Email body is empty , AI returns all nulls, a blank row gets logged in sheet.
- Same order email arrives twice ,Append or Update Row updates existing row instead of creating a duplicate.